

Для начала соберём **zlib** (взято отсюда
<https://github.com/madler/zlib/releases/download/v1.3.2/zlib-1.3.2.tar.xz>):

```
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$ ./configure --prefix="$PWD/_install" --static
Checking for gcc...
Building static library libz.a version 1.3.2 with gcc.
Checking for size_t... Yes.
Checking for off64_t... Yes.
Checking for fseeko... Yes.
Checking for strerror... Yes.
Checking forunistd.h... Yes.
Checking for stdarg.h... Yes.
Checking whether to use vs[n]printf() or s[n]printf()... using vs[n]printf().
Checking for vsnprintf() in stdio.h... Yes.
Checking for return value of vsnprintf()... Yes.
Checking for attribute(visibility) support... Yes.
Checking for s390x build ... No.
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$ make -j$(nproc) CC=arm-linux-gnueabi-hf-gcc
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o example.o test/example.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o adler32.o adler32.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o crc32.o crc32.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o deflate.o deflate.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o infback.o infback.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inffast.o inffast.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inflate.o inflate.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o inftrees.o inftrees.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o trees.o trees.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o zutil.o zutil.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o compress.o compress.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o uncompress.o uncompress.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzclose.o gzclose.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzlib.o gzlib.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzread.o gzread.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -c -o gzwrite.o gzwrite.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -c -o minigzip.o test/minigzip.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o example64.o test/example.c
arm-linux-gnueabi-hf-gcc -O3 -fPIC -D_LARGEFILE64_SOURCE=1 -DHAVE_HIDDEN -I. -D_FILE_OFFSET_BITS=64 -c -o minigzip64.o test/minigzip.c
ar rc libz.a adler32.o crc32.o deflate.o infback.o inffast.o inflate.o inftrees.o trees.o zutil.o compress.o uncompress.o gzclose.o gzlib.o gzread.o gzwrite.o
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$ ls
adler32.c      configure      deflate.o      gzclose.c      gzwrite.o      inflate.c      Makefile      msdos          trees.h        zlib.3          zutil.c
adler32.o      configure.log  doc            gzclose.o      INDEX          inflate.h      Makefile.in   os400          trees.o        zlib.3.pdf      zutil.h
amiga          contrib        example        gzguts.h       infback.c      inflate.o      make_vms.com  qnx            uncompress.o  zlibConfig.cmake.in  zutil.o
BUILD.bazel    crc32.c        example64      gzlib.c        infback.o      inftrees.c    minigzip      README          uncompress.o  zlib.h          zutil.o
Changelog      crc32.h        example64.o    gzlib.o        inffast.c      inftrees.h    minigzip64    README-cmake.md  watcom        zlib.map
CMakeLists.txt  crc32.o        example.o      gzread.c       inffast.h      inftrees.o    minigzip64.o  test           minigzip64.c  zlib.pc
compress.c      deflate.c      examples       gzread.o       inffast.o      libz.a        minigzip.o    treebuild.xml  zconf.h       zlib.pc.cmake.in
compress.o      deflate.h      FAQ            gzwrite.c      inffixed.h     LICENSE        MODULE.bazel  trees.c        zconf.h.in    zlib.pc.in
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$ file trees.o
trees.o: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), not stripped
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$
compress.o      deflate.h      FAQ            gzwrite.c      inffixed.h     LICENSE        MODULE.bazel  trees.c        zconf.h.in    zlib.pc.in
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$ file trees.o
trees.o: ELF 32-bit LSB relocatable, ARM, EABI5 version 1 (SYSV), not stripped
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$ make install
rm -f /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/lib/libz.a
cp libz.a /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/lib
chmod 644 /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/lib/libz.a
rm -f /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/share/man/man3/zlib.3
cp zlib.3 /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/share/man/man3
chmod 644 /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/share/man/man3/zlib.3
rm -f /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
cp zlib.pc /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/lib/pkgconfig
chmod 644 /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/lib/pkgconfig/zlib.pc
rm -f /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/include/zlib.h /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/include/zconf.h
cp zlib.h zconf.h /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/include
chmod 644 /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/include/zlib.h /home/vadim/Repos/OpenSSH/zlib-1.3.2/_install/include/zconf.h
vadim@vadim-PC:~/Repos/OpenSSH/zlib-1.3.2$
```

Далее соберём **OpenSSL** (взято отсюда:
<https://github.com/openssl/openssl/releases/download/openssl-4.0.0/openssl-4.0.0.tar.gz>):

```
vadim@vadim-PC:~/Repos/OpenSSH/openssl-4.0.0$ ./Configure --cross-compile-prefix=arm-linux-gnueabi-hf- --prefix="$PWD/_install" linux-generic32
Configuring OpenSSL version 4.0.0 for target linux-generic32
Using os-specific seed configuration
Created configdata.pm
Running configdata.pm
Created Makefile.in
Created Makefile
Created include/openssl/configuration.h

*****
***
***   OpenSSL has been successfully configured
***
***   If you encounter a problem while building, please open an
***   issue on GitHub <https://github.com/openssl/openssl/issues>
***   and include the output from the following command:
***
***       perl configdata.pm --dump
***
***   (If you are new to OpenSSL, you might want to consult the
***   'Troubleshooting' section in the INSTALL.md file first)
***
*****
vadim@vadim-PC:~/Repos/OpenSSH/openssl-4.0.0$ make install
make install_sw
make[1]: Entering directory '/home/vadim/Repos/OpenSSH/openssl-4.0.0'
/usr/bin/perl util/mkinstallvars.pl COMMENT="This file should be used when building against this OpenSSL build, and should never be installed" P
REFIX= _BINDIR=apps APPLINKDIR=ms LIBDIR= INCLUDEDIR=include "INCLUDEDIR=/include" MODULESDIR=providers libdir= CMAKECONFIGDIR= PKGCONFIGDIR= "
```

```
install doc/html/man7/provider-skeymgmt.html -> /home/vadim/Repos/OpenSSH/openssl-4.0.0/_install/share/doc/openssl/html
install doc/html/man7/provider-storemgmt.html -> /home/vadim/Repos/OpenSSH/openssl-4.0.0/_install/share/doc/openssl/html
install doc/html/man7/provider.html -> /home/vadim/Repos/OpenSSH/openssl-4.0.0/_install/share/doc/openssl/html/man7
install doc/html/man7/proxy-certificates.html -> /home/vadim/Repos/OpenSSH/openssl-4.0.0/_install/share/doc/openssl/html/man7
install doc/html/man7/x509.html -> /home/vadim/Repos/OpenSSH/openssl-4.0.0/_install/share/doc/openssl/html/man7
make[1]: Leaving directory '/home/vadim/Repos/OpenSSH/openssl-4.0.0'
vadim@vadim-PC:~/Repos/OpenSSH/openssl-4.0.0$ ls _install
bin include lib share ssl
vadim@vadim-PC:~/Repos/OpenSSH/openssl-4.0.0$ ls _install/bin
openssl
vadim@vadim-PC:~/Repos/OpenSSH/openssl-4.0.0$ ls _install/lib
cmake libcrypto.a libcrypto.so libcrypto.so.4 libssl.a libssl.so libssl.so.4 ossl-modules pkgconfig
vadim@vadim-PC:~/Repos/OpenSSH/openssl-4.0.0$
```

Теперь собираем **OpenSSH** (взято отсюда

https://github.com/openssh/openssh-portable/archive/refs/tags/V_10_3_P1.tar.gz).

Предварительно сконфигурируем сборку так, чтобы не вызывался strip, в противном случае make попытается применить нативный в системе strip, а не arm-linux-gnueabi-hf-strip:

```
vadim@vadim-PC:~/Repos/OpenSSH/openssh-portable-V_10_3_P1$ ./configure --prefix=
"$PWD/_install" --host=arm-linux-gnueabi-hf --with-zlib="$PWD/../zlib-1.3.2/_install" --with-ssl-dir="$PWD/../openssl-4.0.0/_install" --with-privsep-path="$PWD/empty" --disable-strip
checking for arm-linux-gnueabi-hf-cc... no
checking for arm-linux-gnueabi-hf-gcc... arm-linux-gnueabi-hf-gcc
checking whether the C compiler works... yes
checking for C compiler default output file name... a.out
checking for suffix of executables...
checking whether we are cross compiling... yes
```

```
vadim@vadim-PC:~/Repos/OpenSSH/openssh-portable-V_10_3_P1$ make install-nosysconf
Makefile:730: warning: ignoring prerequisites on suffix rule definition
conffile=`echo sshd_config.out | sed 's/.out$/.'; \
/usr/bin/sed -e 's|/etc/ssh/ssh_config|/home/vadim/Repos/OpenSSH/openssh-portable-V_10_3_P1/_install/etc/ssh_config|g' -e 's|/etc/ssh/ssh_known_hosts|/home/vadim/Repos/OpenSSH/openssh-portable-V_10_3_P1/_install/etc/ssh_known_hosts|g' -e 's|
```

```
portable-V_10_3_P1/_install/share/man/man8/ssh-sk-helper.8
vadim@vadim-PC:~/Repos/OpenSSH/openssh-portable-V_10_3_P1$ ls _install
bin libexec sbin share
vadim@vadim-PC:~/Repos/OpenSSH/openssh-portable-V_10_3_P1$ ls _install/sbin
sshd
vadim@vadim-PC:~/Repos/OpenSSH/openssh-portable-V_10_3_P1$
```

Всё собрано, осталось сформировать **initramfs**.

Для начала возьмём из **/usr/arm-linux-gnueabi-hf/lib** библиотеки **ld-linux-armhf.so.3** и **libc.so.6**, после этого заодно перенесём динамические библиотеки **libcrypto.so.4** и **libssl.so.4**:

```
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs/lib$ ls -l
total 6832
-rwxr-xr-x 1 vadim vadim 1074636 Apr 15 2024 ld-linux-armhf.so.3
lrwxrwxrwx 1 vadim vadim 14 Jun 10 15:03 libcrypto.so -> libcrypto.so.4
-rwxr-xr-x 1 vadim vadim 3885188 Jun 10 14:24 libcrypto.so.4
-rwxr-xr-x 1 vadim vadim 1114352 Apr 15 2024 libc.so.6
lrwxrwxrwx 1 vadim vadim 11 Jun 10 15:03 libssl.so -> libssl.so.4
-rwxr-xr-x 1 vadim vadim 903528 Jun 10 14:24 libssl.so.4
drwxrwxr-x 2 vadim vadim 4096 Jun 10 14:24 ossl-modules
drwxrwxr-x 2 vadim vadim 4096 Jun 10 14:24 pkgconfig
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs/lib$
```


Добавляем бинарники из OpenSSH:

```
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs$ ls -ld bin/*ssh*
-rwxr-xr-x 1 vadim vadim 3146788 Jun 10 14:39 bin/ssh
-rwxr-xr-x 1 vadim vadim 1093684 Jun 10 14:39 bin/ssh-add
-rwxr-xr-x 1 vadim vadim 950692 Jun 10 14:39 bin/ssh-agent
-rwxr-xr-x 1 vadim vadim 1428592 Jun 10 14:39 bin/ssh-keygen
-rwxr-xr-x 1 vadim vadim 1914052 Jun 10 14:39 bin/ssh-keyscan
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs$ ls -ld sbin/*ssh*
-rwxr-xr-x 1 vadim vadim 1451716 Jun 10 14:39 sbin/sshd
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs$
```

Добавим конфигурацию системы в /etc:

```
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs/etc$ openssl passwd -salt 123
-6 12345
$6$123$j27M/oYEBKELS7uKr0EzyLXT//fhdqpNk9iUmgkiTaBKB4EQQK4VXdXfaU3fLvGEpNmatQUeA
LZ8J45U.43cI0
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs/etc$ cat passwd
root:x:0:0:root:/root:/bin/ash
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs/etc$ cat shadow
root:$6$123$j27M/oYEBKELS7uKr0EzyLXT//fhdqpNk9iUmgkiTaBKB4EQQK4VXdXfaU3fLvGEpNma
tQUeALZ8J45U.43cI0:20080:0:99999:7:::
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs/etc$ cat ssh/sshd_config
Port 12345
PubkeyAuthentication no
PasswordAuthentication yes
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs/etc$
```

Упаковываем initramfs:

```
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs$ find . | grep -v './initramfs
.cpio.gz' | cpio -o -H newc | gzip > initramfs.cpio.gz
60806 blocks
vadim@vadim-PC:~/Emulation/arm-vexpress/initramfs$
```

Запускаем прошивку:

```
vadim@vadim-PC:~/Emulation/arm-vexpress$ QEMU_AUDIO_DRV=none qemu-system-arm -M vexpress-a9 -kerne
l zImage -initrd initramfs.cpio.gz -dtb vexpress-v2p-ca9.dtb -append "console=ttyAMA0 rdinit=/bin/
ash" -nographic
[ 0.000000] Booting Linux on physical CPU 0x0
[ 0.000000] Linux version 6.18.33 (vadim@vadim-PC) (arm-linux-gnueabi-gcc (Ubuntu 13.3.0-6ubu
ntu2~24.04.1) 13.3.0, GNU ld (GNU Binutils for Ubuntu) 2.42) #1 SMP Fri May 29 15:29:39 +07 2026
[ 0.000000] CPU: ARMv7 Processor [410fc090] revision 0 (ARMv7), cr=10c5387d
[ 0.000000] CPU: PIPT / VIPT nonaliasing data cache, VIPT nonaliasing instruction cache
[ 0.000000] OF: fdt: Machine model: V2P-CA9
```

Проверяем работоспособность бинарников из OpenSSH:

```

ls
bin      etc      libexec  root      ssl
dev      lib      linuxrc  sbin      usr
~ # [ 14.577947] amba 1000f000.wdt: deferred probe pending: (reason unknown)
[ 14.578067] amba 100e0000.memory-controller: deferred probe pending: (reason unknown)
[ 14.578089] amba 100e1000.memory-controller: deferred probe pending: (reason unknown)
[ 14.578103] amba 100e5000.watchdog: deferred probe pending: (reason unknown)
cd root
/root # ssh-keygen
Couldn't open /dev/null: No such file or directory
/root # mknod /dev/null c 1 3
/root # chmod 666 /dev/null
/root # ssh-keygen
[ 355.836237] random: crng init done
Generating public/private ed25519 key pair.
Enter file in which to save the key (/root/.ssh/id_ed25519): my_key
Enter passphrase for "my_key" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in my_key
Your public key has been saved in my_key.pub
The key fingerprint is:
SHA256:u5T7096j2CJPQxiTwJZhEbZmaVA37U+kmED7y4cZ0T0 root@(none)
The key's randomart image is:
+--[ED25519 256]--+
|      .+X=o.      |
|      +=* +...    |
|      .B.=+.oE    |
|      + .o=o ..   |
|      S .o        |
|      . O .       |
|      O +.        |
|      ..=o+...    |
|      o++++...    |
+-----[SHA256]-----+
/root # cat my_key.pub
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIFJz6nZLrODtczQF9jn8uS1GwiD9NPd/kGWIwG2dcCNr root@(none)
/root #

```

```

/root # sshd
sshd requires execution with an absolute path
/root # /sbin/sshd
/home/vadim/Repos/OpenSSH/openssh-portable-V_10_3_P1/_install/etc/sshd_config: No such file or directory
/root # /sbin/sshd -f /etc/ssh/sshd_config
sshd: no hostkeys available -- exiting.
/root #

```

```

/root # openssl passwd -6 My_Password
$6$.iSvZkL7AT737eQs$4VhmyLNJRig1kdjWCvz5oQXmVZW0/TeX5bdcWQct8t4N59sydZntoXJStfPrnCubxbawKmB3Y0xjIDWGyk81
/root # openssl dgst -sha512 my_key.pub
SHA2-512(my_key.pub)= 27f4d0250951a1521720bceec0a2f3420313449637df109d22ec07a7a13e6085f57c4daa3182ee840a233adb9ddd34a39f17485636d814721709307d60f1bed5
/root #

```